Your grade: 100%

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Next item →

1. What is the primary goal of Principal Components Analysis (PCA)? 1 / 1 point

- To minimize the dimensionality of the dataset while maximizing data loss
- To classify the data into distinct categories based on variance
- To compute a new set of features that preserves information about variance in the original dataset.

Nice work

Correct. PCA aims to derive a new set of features from the original dataset by taking linear combinations of the original dataset columns, in a way that preserves the variance information but in a more efficient manner.

2. Why is it important to always normalize data before applying PCA? 1 / 1 point

- To reduce computational complexity
- To ensure each feature contributes equally
- To find the optimal number of principal components
- To simplify the interpretation of components

Nice work

Normalizing the data before applying PCA is crucial because it scales all features to a uniform range. This prevents features with larger scales from dominating the variance calculations, ensuring that each feature contributes equally to the analysis.

3. In the context of PCA, what do the resulting 'loadings' represent? 1 / 1 point

- Correlation coefficients
- Weights of the linear combinations assigned to each original feature
- Eigenvalues of the covariance matrix
- Scaled data points

Nice work

In PCA, loadings refer to the weights assigned to each original feature in the linear combinations that form the new features, reflecting their contribution to each principal component.

4. In a study analyzing the features of tumor cells, a researcher using a biplot after performing PCA on the data, in which more than 80% of the variance in the data is captured by the first two principal components. If two variables 'mean texture' and 'mean perimeter' have vectors in the biplot that only have a small angle between them, what can be inferred about their relationship? 1 / 1 point

- They are highly correlated
- They have a weak correlation.
- Both variables likely represent noise data, and so can be removed from the analysis

Nice work

In a biplot, the cosine of the angle between two variable markers (vectors) indicates the coefficient of correlation between those variables. Vectors that have a small angle between them have a cosine of the angle close to 1, indicating a high correlation.

5. What does a biplot display in PCA analysis? 1 / 1 point

- A histogram of principal component values superimposed on a data distribution chart in a single plot
- The eigenvalues and eigenvectors of the resulting PCA transform
- Both the scores (transformed data) and the loadings (relation between the principal components and the data)

Nice work

Biplots are useful in PCA analysis as they show both the transformed data (scores) and the relationships between the principal components and the data (loadings) in one plot.

6. Which statement about Principal Component Analysis is *not* correct? 1 / 1 point

- The values in a principal component vector are called 'loadings'.
- Any given data matrix can have at most two principal components
- Each principal component vector represents the weights for a particular linear combination of the original dataset columns.
- The coordinates of an original data point, when transformed into the feature space formed by principal components, are called 'scores'.

Nice work

False: in PCA, the maximum number of principal components is equal to the number of original variables (columns) in the dataset.